

Asal Rabiee

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RESEARCH INTERESTS

Computational Genomics and Transcriptomics, Systems Biology and Gene Regulatory Networks, Non-coding RNA and Gene Regulation, Machine Learning for Biological Data, Computational Drug Discovery and Structure-Based Modeling

EDUCATION

B.Sc. in Animal Biology (Concentration: Bioinformatics)

Sep 2021- Feb 2026

University of Tehran - GPA: 18.5/20 (3.9/4.00)

- Coursework: Systems Biology, Epigenetics and Gene Regulation, Molecular Genetics, Biostatistics, Cell and Molecular Biology, Structural Biochemistry, Genetic Engineering and Biotechnology
- Bachelor thesis: Benchmarking Protein-Ligand Binding Site Prediction with Pseq2Sites, [GitHub](#)
- Supervisor: Dr. Najmeh Salehi

Minor in Computer Science

Feb 2022- Feb 2026

University of Tehran

- Coursework: Bioinformatics, Artificial Intelligence, Data Structures & Algorithms, Database Systems

PUBLICATIONS

- Farahzadeh, G., Rezaeirad, A., Hassani, S., Farajzadeh Tarani, M., Peymankar, A., **Rabiee, A.**, & Roohparvar Basmenj, E. (2024). QSAR, docking and molecular dynamic simulation studies of leishmanial inhibitors using a bioinformatics approach. *Authorea*. <https://doi.org/10.22541/au.172464590.09500449/v1>
- **Rabiee, A.**, & Habibi-Rezaei, M. (2023). Deciphering the mechanisms of blood sugar homeostasis: The story of Gerty and Carl Cori. *Science Cultivation*. https://www.sciencecultivation.ir/article_705829.html

RESEARCH EXPERIENCE

School of Biology, College of Science, University of Tehran

Sep 2026- Present

- **Structure-Based Drug Discovery Targeting PAD2:** Investigating small-molecule allosteric inhibitors of PAD2 through protein and ligand preparation, binding-site characterization, molecular docking and redocking validation, and structure-based virtual screening using Schrödinger Maestro.

School of Biology, College of Science, University of Tehran

Jan 2025- Present

- **Protein-Ligand Binding Site Prediction (Pseq2Sites):** Benchmarked a CNN + attention model for sequence-based protein-ligand binding site prediction using scPDB and PDBbind datasets and ProtTrans embeddings; evaluated model feasibility, limitations, and potential methodological improvements. [GitHub](#)
- **lncRNA-Ion Channel Relationships in Gastric Cancer:** Analyzed TCGA-STAD RNA-seq data using differential expression analysis and WGCNA to identify lncRNA-ion channel co-expression modules associated with clinical traits and prioritize candidate genes for further investigation. (Manuscript in preparation.) [GitHub](#)

School of Biology, College of Science, University of Tehran

Sep 2023- Aug 2024

- **Computational Drug Discovery for Leishmaniasis:** Collaborated in a seven-member research team with researchers from IAU, Uskudar University, Semnan University, SUT, KMU, and TMU on bioinformatics-driven drug discovery for leishmaniasis. Contributed to literature review and computational analyses, supporting the integration of QSAR modeling, molecular docking, and molecular dynamics simulations. (Published as preprint)

Protein Biotechnology Research Laboratory (PBRL), University of Tehran

Sep 2022- Dec 2023

- **Cannabidiol and the Endocannabinoid System in Alzheimer's Disease:** Explored the therapeutic potential of Cannabidiol (CBD) via modulation of the Endocannabinoid System (ECS) in Alzheimer's disease, focusing on neuroprotection and synaptic regulation.
- **Mechanisms of Blood Glucose Homeostasis:** Authored a scholarly review on the historical and mechanistic exploration of blood sugar homeostasis through the pioneering work of Nobel laureates Gerty and Carl Cori. (Published in internal scientific journal)

TEACHING EXPERIENCE

Teaching Assistant

Jan 2023- Present

School of Biology and Computer Science, College of Science, University of Tehran

- Assisted in teaching undergraduate courses (60+ students), supporting instruction, grading, and exam supervision, while strengthening **communication**, **teamwork**, and **leadership** skills.
- Courses: **Systems Biology, Bioinformatics, Biostatistics, Molecular Genetics, Metabolic Biochemistry, Structural Biochemistry, Neurophysiology & Endocrinology**

Laboratory Assistant

Sep 2024- Jul 2025

School of Biology, College of Science, University of Tehran

- Assisted in two large laboratory courses (80+ students), supporting experiments, data analysis, and student guidance. Strengthened **technical**, **mentoring**, and **lab management** skills.
- Courses: **Genetics Laboratory** - *Drosophila* genetics, blood typing, pedigree analysis, Barr body detection, polytene chromosome observation. **Molecular Genetics Laboratory** - DNA extraction, quantification (Nanodrop), primer design (Oligo7), PCR, electrophoresis, ORF detection (bioinformatics project).

SKILLS

Bioinformatics:	RNA-seq Analysis, Systems Biology (WGCNA), Sequence Alignment (Clustal Omega, BLAST+), Reference Mapping (BWA, SAMtools), Genome Assembly (SPAdes, Quast)
Programming:	Python, R, Bash, SQL, C++, $\LaTeX$
Machine Learning:	PyTorch, Scikit-Learn, TensorFlow/Keras, Transformers
Software Tools:	Cytoscape, IGV, UCSF Chimera, Open Babel, SnapGene, Oligo7, GenoPro, SPSS, EndNote
Version Control:	Git, GitHub
Molecular Modeling:	QSAR, Molecular Docking (AutoDock, Schrödinger Maestro), Molecular Dynamics (GROMACS)
Laboratory:	PCR and Primer Design, DNA Electrophoresis, Genomic DNA Extraction (Salting-out), Spectrophotometry, Nanodrop, Centrifugation, Blood Smear Preparation, ELISA, Aseptic Technique

COURSEWORK PROJECTS

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- Identification of Xylanase Genes:** Developed a bioinformatics pipeline to identify thermostable xylanase genes from rumen metagenomes using BLAST+, CD-HIT, Clustal Omega, and HMMER for sequence screening, clustering, alignment, and profile-based identification. Curated candidate sequences to generate a high-confidence gene set. [GitHub](#)
 - Read Mapping and Genome Assembly:** Implemented an NGS workflow on *E. coli* short-read data: quality control (FastQC), de novo assembly (SPAdes, Quast), and read mapping (BWA, SAMtools). Validated alignments in IGV and evaluated concordance, mapping rates, and read depth. [GitHub](#)
 - Malaria Detection Using Convolutional Neural Networks:** Developed a CNN-based image classification model to distinguish parasitized and uninfected red blood cells using microscopy images. Implemented data preprocessing, regularization (dropout, batch normalization), and performance evaluation for robust binary classification. [GitHub](#)
 - Hierarchical Data Structure for Genetic Diseases:** Implemented a tree-based data structure for efficient representation and querying of hierarchical genetic disease relationships. [GitHub](#)
 - Genetic Database:** Designed and implemented a BCNF-normalized relational database in PostgreSQL with ER models, schemas, and role-based access control. [GitHub](#)

HONORS

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- Ranked 1st in Cumulative GPA among undergraduate students in my major, University of Tehran, 2025.
 - Awarded a full scholarship for the Computer Science minor, University of Tehran, 2022.
 - Awarded a full scholarship for undergraduate studies, University of Tehran, 2021.
 - Top 8.5% in National University Entrance Exam among more than 500,000 participants, Tehran, 2021.

CONFERENCES ATTENDED

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- Attended the Healthcare Bioinformatics and Computational Biology (HBC2025) Conference, Online, February 2025. [Certificate of Participation](#)

EXTRACURRICULAR ACTIVITIES

Volunteer Tour Guide

Sep 2022- Apr 2024

Zoological Museum, University of Tehran

- Led interactive museum tours for diverse audiences, introducing concepts in general biology, evolution, and biodiversity while communicating complex biological topics in an accessible and engaging manner.

REFERENCES

References available upon request.